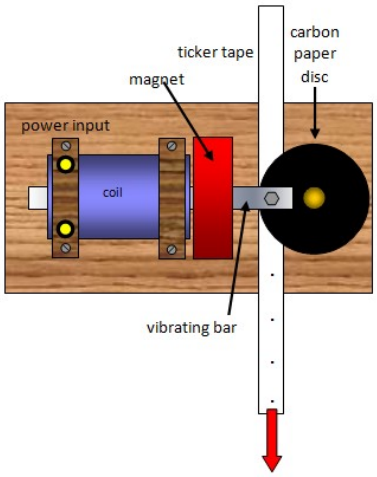
Year 10 Physics
Measuring Velocity Using the Ticker Timer

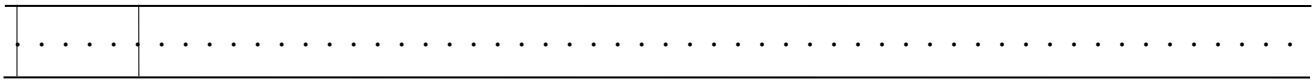
Name: _____

A ticker timer is an instrument that can be used to measure velocity. To measure we need to measure t_____ and _____. A ticker timer consists of a ticker tape that is attached to the moving object. The ticker tape passes below a metal bar that vibrates 50 times per second making dots (i.e. 50 dots per second). Therefore the time between each dot is 1/50 s (or 0.02 s). To Measure distance we measure the length between dots. To make it simpler we will measure 5 dots at a time which is 0.01 s (i.e. 5 × 0.02= 0.1s).



$$\text{Velocity} = \frac{distance\,(cm)}{time\,(s)}$$

Ticker Timer 1



Complete the following table for TT 1

No. dots	Time (s)	Total time (s)	Distance (cm)	Total distance (cm)	Velocity = $\frac{distance\,(cm)}{time\,(s)}$
5	0.1	0.1	1.6	1.6	
5	0.1	0.2	1.6	3.2	
5	0.1	0.3	1.6	4.8	
5	0.1	0.4	1.6	6.4	
5	0.1	0.5	1.6	8.0	
5	0.1	0.6	1.6	9.6	
5	0.1	0.7	1.6	11.2	

Questions

1. Describe the pattern of the dots in TT1?
2. What was the velocity after the first 5 dots?
3. What was the velocity after the last 5 dots?
4. If this was a car on the road, what would you notice about it speed if you were in a helicopter above?

_____

Ticker Timer 2

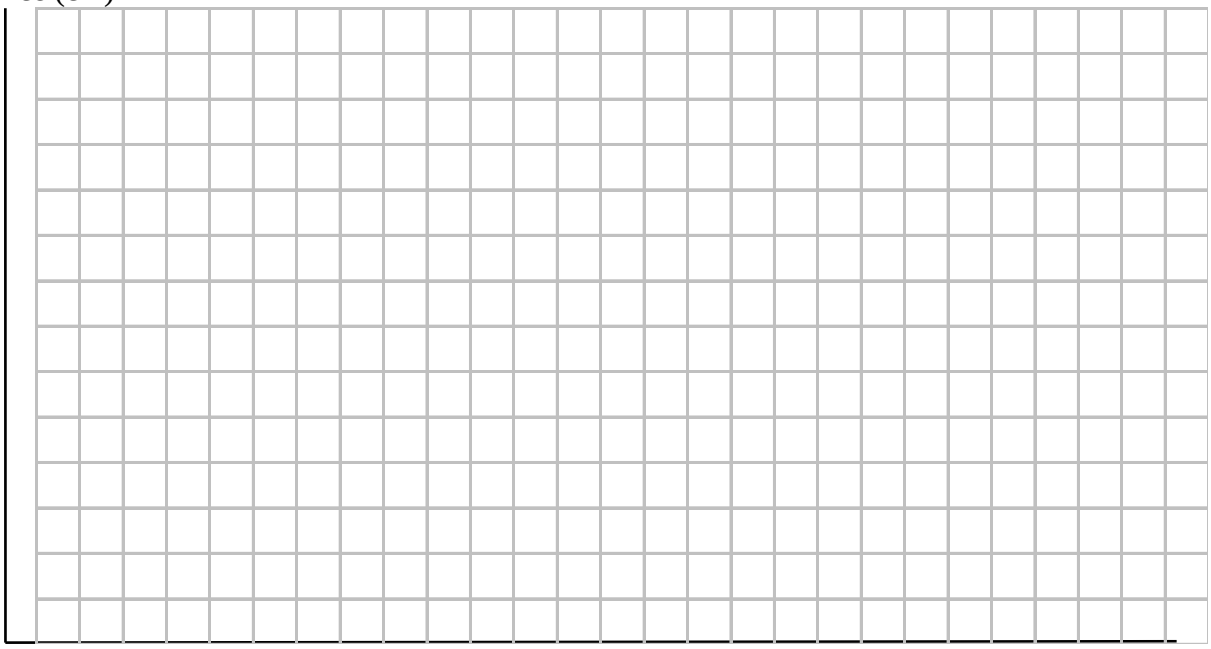
Complete the following table for TT 2

No. dots	**Time (s)**	**Total time (s)**	**Distance (cm)**	**Total distance (cm)**	Velocity = $\frac{distance\ (cm)}{time\ (s)}$	Change in velocity $v_2 - v_1$
5	0.1	0.1				
5	0.1	0.2				
5	0.1	0.3				

Questions

1. Describe the pattern of the dots in TT2?
2. What was the velocity after the first 5 dots?
3. What was the velocity after the last 5 dots?
4. If this was a car on the road, what would you notice about it speed if you were in a helicopter above?
5. In TT2 we say the car was accelerating what do we mean by that, in terms of velocity?

Distance (cm)



Time (s)

